Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/06/2007
PI Name:	Grotberg, James M.D., Ph.D.		
Project Title:	Fluid Dynamics and Interfacial Stability of Pulmonary Airway Closure and Reopening in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	03/12/2002	**End Date:**	02/12/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 2/12/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2740		
Performance Goal No.:			
Performance Goal Text:			

Our research intends to study multiphase flow behavior related to human lungs in microgravity by both theoretical and experimental means. Since the lung's airways are internally coated with a thin liquid film, closure of the airway can occur by a capillary instability resulting in formation of a liquid plug. This is most often found near the end of expiration, but sooner under various abnormal conditions. A closed airway cannot participate in gas exchange. In normal gravity, closure is limited to the lower lung due to the body force on the tissues, while the upper lung remains open. In microgravity, however, it can be more uniformly distributed. Once closed we are interested in how to reopen the airway. This process involves propagation of the formed liquid plug distally during an inspiration. As the plug deposits its contents on the airway walls, it can eventually rupture and gas exchange is again possible.

Our project examines the dynamics of the capillary instability responsible for airway closure and how potentially one could prevent closure from occurring. The system is mathematically modeled and solved using a combination of asymptotic and numerical methods. The role of fluid properties, wall elasticity, surface charges, imposed oscillations and surfactants in potential stabilization of the closure process will be assessed. In addition, the process of airway reopening due to the dynamics of plug propagation will be examined through a mathematical model of two-phase flow in a tube with a similar parameter variation as the closure problem. A sequence of experiments is proposed to evaluate the effects of modifying the system parameters.

Task Description:

The closure and subsequent reopening of lung airways due to capillary-elastic instabilities contributes strongly to the mechanical response of the lung during breathing. Understanding the underlying fluid mechanical phenomena is a fundamental inquiry in lung physiology and function, both in microgravity situations and on earth.

The liquid plug propagation which occurs in airway reopening is also a phenomenon seen in other pulmonary applications. For example, treatment of certain diseases involves instillation of drugs mixed with liquid directly into the trachea and forced throughout the lung by imposed ventilation. This is the common mode of delivering surfactants into the lungs of prematurely-born infants who suffer from surfactant deficiency (hyaline membrane disease, also called respiratory distress syndrome of the newborn). Liquid plugs are formed during these procedures and blown distally by mechanical inspiration. The resulting spatial distribution in the lung depends on the system parameters governing the basic plug flow phenomenon that we study. Delivery of genetic material and other drugs into the lung can be done by liquid instillation techniques, too.

Another pulmonary example is liquid ventilation, a means of ventilating a critically ill patient with perfluorocarbon liquid. The process of filling the lung with the perfluorocarbon, as well as the development and motion of plugs are essential to the effectiveness of the treatment and depend on the liquid plug flows we are examining. In addition to the pulmonary application, this research addresses issues which are basic to other engineering fields. In microfluidic devices which involve gas and liquid flows, it is common to have plug propagation as an underlying means of moving samples through the chip. The length scales for these devices is quite similar to the small airways of the lung, so our results are directly applicable to this rapidly growing field. In manufacturing there are two air-liquid free surfaces in annular extrusion methods for creating small hollow fibers. These may be materials processed for a number of applications. Also, there are methods of co-extrusion manufacturing (e.g. optical fibers) in which there is an inner plug core of liquid and an outer annulus of a different liquid extruded simultaneously, one around the other. Also, these extrusions do not remain entirely fluid. Even though initially both surfaces in these applications are fluid-fluid interfaces, the outer surface in extruded forms will cool before the inner one, and that surface will develop flexible, elastic properties during part of the cooling process.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

In Fujioka and Grotberg[1] we investigate the steady propagation of a liquid plug within a two-dimensional channel lined by a uniform, thin liquid film. For a fixed plug length, we find that the trailing film thickness increases with plug propagation speed. As the plug length decreases below the channel width, the trailing film thickness decreases, and for finite Re there is significant interaction between the leading and trailing menisci. The macroscopic pressure gradient, i.e., the pressure drop between the leading and trailing gas phases divided by the plug length, is a function of U and U^2, where U is the plug propagation speed. A capillary wave develops at the front meniscus, with an amplitude that increases with Re, and this causes large local changes in wall shear stresses and pressures.
In Wei, Fujioka, Hirschl and Grotberg[2] an alveolus partially filled with liquid in the presence of surfactants is modeled. The presence of surfactants induces a nontrivial cycle-averaged surfactant concentration gradient along the interface that generates steady streaming. The steady streaming patterns (e.g., number of vortices) depend on the ratio of inspiration to expiration periods and the sorption parameter. For an insoluble surfactant, a single vortex is always formed. However, with a soluble surfactant, more complex flow patterns can appear such as three vortices or saddle-point flow structures. We also demonstrate that convection is either comparable to or more dominant than diffusion, and this is particularly important e.g. for a cell-cell signaling molecule such as surfactant-associated protein-A for regulating surfactant secretion.

In Wei, Halpern and Grotberg[3] we analyze the effect of an insoluble surfactant on the stability of an annular film in a capillary undergoing a time-periodic pressure gradient force. An analysis for thin films is used to derive a coupled set of equations for the perturbed fluid-fluid interface and the interfacial surfactant concentration. Under certain conditions the interaction between a surfactant and a steady base flow may induce a faster growth of the disturbances of the interface than a stationary base state. Time-periodic base flows can modify the features of the steady-flow-based instability, depending on surface tension, surfactant activity, and oscillatory frequency. For an oscillatory base flow with zero mean, the growth rate decreases with increasing frequency. In the low-frequency limit, the growth rate approaches a maximum corresponding to the growth rate of a steady base flow having the same amplitude. In the high-frequency limit, the growth rate reaches a minimum corresponding to the growth rate in the limit of a stationary base state.

In Suresh and Grotberg[4] the effect of plug propagation speed and gravity on the quasisteady motion of a liquid plug in a two-dimensional liquid-lined channel oriented at an angle alpha with respect to gravity is studied. It is assumed that the capillary number Ca is small while the Bond number Bo is arbitrary. The thickness of the trailing films left behind by the plug and the pressure drop across it are determined as functions of Ca, Bo and alpha and the thickness of the precursor films. When Bo is small, the trailing film thickness and the flow contribution to the pressure drop scale as Ca^2/3 with coefficients that depend on Bo and alpha. By computing V_R, the ratio of liquid volumes above and below

the centerline of the channel, the asymmetry in the liquid distribution can be quantified. V_R decreases with Bo and Ca, but increases with plug length L_p. The decrease of V_R with Ca suggests that higher propagation speeds in small airways may result in less homogenous liquid distribution, which is in contrast to the expected effect in large airways. For given values of the other parameters, a maximum capillary number Ca_c is identified above which the plug will rupture. When becomes equal to an orientation-dependent critical value Bo_c, the film thickness and pressure drop scalings change to Ca^1/2 and Ca^1/6, respectively.

In Fujioka and Grotberg[5] we investigate the steady propagation of a liquid plug containing soluble surfactant in a two-dimensional channel lined by a uniform, thin liquid film. As the plug propagates forward, surfactant accumulates on the front meniscus interface. The surfactant concentration is a maximum somewhere in the front meniscus, and a surface tension gradient is established that opposes flow out of the film region. In this region, the surface velocity almost vanishes, and this results in the precursor film thickness near the meniscus being thicker than the leading film thickness. Because of an increase in the minimum film thickness, there is a reduction in the peaks of wall pressure and shear stress. However, the overall pressure drop across the plug increases as the result of the increasing surfactant concentration. A recirculation flow forms inside the plug core, which becomes more skewed toward the rear meniscus as the Reynolds number increases. As the surfactant concentration increases, there is rigidifcation of the front interface, and the recirculation flow moves away from the front meniscus. Subsequently, the recirculation flow is directed away from the rear interface. When the plug length is shorter, this change in recirculation pattern occurs at a smaller surfactant concentration.

In Zheng, Anderson, Suresh and Grotberg[6] a model of a symmetrically bifurcating airway was used to investigate the effect of gravity on the distribution of instilled liquids. A liquid plug was instilled into the parent tube and driven through the bifurcation by a syringe pump. The influence of gravity was varied by adjusting the roll and pitch angles phi and gamma of the bifurcation. An important quantity is the splitting ratio R_s of the liquid volume in the upper daughter to the lower just after plug splitting. It is a function of the parent-tube capillary number Ca_p, the Bond number Bo, the gravitational angles, and the presence of pre-existing plugs initially blocking either daughter tube. Below a critical capillary number Ca_c no liquid entered the upper daughter, and above which R_s increased and leveled off with Ca_p. More liquid entered the unblocked daughter with a blockage in one daughter tube, and this effect was larger with larger gravitational effect. A theoretical model that predicts R_s and Ca_c is shown to be in qualitative agreement with the experiments over a wide range of parameters.

In Zheng, Fujioka, Grotberg and Grotberg[7] and [9], the effects of inertia and gravity on liquid plug splitting in a bifurcation model to simulate the liquid distributions in large airways are investigated by conducting experiments and theoretical studies. The splitting ratio R_s is measured as a function of parent tube Reynolds number Re_p, roll and pitch angles, the plug length and the presence of pre-existing plug blockages in downstream tubes. Results show that increasing causes more homogeneous splitting, and that there is a critical such that no liquid enters the upper daughter tube. increases while R_s decreases with increasing gravitational angles. Inertia is demonstrated to cause more homogeneous plug splitting. A theoretical model based on entrance flow for the plug in the daughters is developed from which R_s and the frictional pressure drop can be computed as function of Re_p. This theory provides a good prediction on liquid plug splitting and liquid distribution in the large airways of human lungs.

Task Progress:

In Zheng, Fujioka and Grotberg[8], [12], [14] and [15] a model of steady plug propagation driven by gravity and pressure in a 2-D liquid-lined channel oriented at an angle alpha with respect to gravity is developed. The important parameters are Bo, alpha, Ca, Re, L_p and the surfactant concentration C_0. If gravity is neglected the plug is symmetric, there is no flow interaction between the upper and lower half plug domains, and two flow regimes exist: two wall layers and two trapped vortices in the core. The number of vortices is 0, 1 or 2 when Bo and alpha are nonzero, as fluid flows from the upper precursor film, through the core and into the lower trailing film. For C_0>0 the vortices detach from the interface and create saddle points inside the core. A capillary surface wave develops on the front meniscus that extends into the precursor film. In this region the film is thinnest and the wall shear stress attains its maximum value, which could be large enough to cause epithelial cell damage. The peak magnitude of the shear stress can be reduced by the addition of surfactant.

In Fujioka, Halpern and Grotberg [16] and [18], we extended our previous airway closure models by considering the effects of a shear thinning liquid layer coating the inner surface of a rigid cylindrical tube and an oscillatory air-core phase. The presence of the core flow enhances the initial film growth-rate when compared to the no core flow case. As the liquid bulge grows, its axial displacement increases due to the oscillatory core flow. At some point, the film growth-rate decreases and eventually the minimum core radius approaches a non-zero value implying that a liquid plug has not formed. The effect of core flow frequency and amplitude and properties of the film fluid are investigated.

Refereed Journal Publications (quoted above)

1. Fujioka, H. and J.B. Grotberg, Steady propagation of a liquid plug in a two-dimensional channel. Journal of Biomechanical Engineering-Transactions of the Asme, 2004. 126(5): p. 567-577.

2. Wei, H.H., H. Fujioka, R.B. Hirschl, and J.B. Grotberg, A model of flow and surfactant transport in an oscillatory alveolus partially filled with liquid. Physics of Fluids, 2005. 17(3).

3. Wei, H.H., D. Halpern, and J.B. Grotberg, Linear stability of a surfactant-laden annular film in a time-periodic pressure-driven flow through a capillary. Journal of Colloid and Interface Science, 2005. 285(2): p. 769-780.

4. Suresh, V. and J.B. Grotberg, The effect of gravity on liquid plug propagation in a two-dimensional channel. Physics of Fluids, 2005. 17(3).

5. Fujioka, H. and J.B. Grotberg, The steady propagation of a surfactant-laden liquid plug in a two-dimensional channel. Physics of Fluids, 2005. 17(8).

6. Zheng, Y., J.C. Anderson, V. Suresh, and J.B. Grotberg, Effect of gravity on liquid plug transport through an airway bifurcation model. Journal of Biomechanical Engineering-Transactions of the Asme, 2005. 127(5): p. 798-806.

7. Zheng, Y., H. Fujioka, J.C. Grotberg, and J.B. Grotberg, Effects of inertia and gravity on liquid plug splitting at a bifurcation. Journal of Biomechanical Engineering-Transactions of the Asme, 2006. 128(5): p. 707-716.

8. Zheng, Y., H. Fujioka, and J.B. Grotberg, Effects of gravity, inertia and surfactant on steady plug propagation in a two-dimensional channel. Physics of Fluids, 2007.

	Abstracts and presentations (quoted above) 9. Zheng, Y., and J.B. Grotberg. Effect of inertia and gravity on liquid plug splitting at a bifurcation. Biomedical Engineering Society Annual Fall Meeting, Philadelphia, PA, October, 2004. 10. Fujioka, H., and J.B. Grotberg. The oscillatory motion of a surfactant-laden liquid plug in a 2D-channel. American Physical Society, Division of Fluid Dynamics, 57th Annual Meeting, Seattle, WA, November, 2004. 11. Fujioka, H. and J.B. Grotberg. Effect of Surfactant on Wall Stresses During Liquid Plug Propagation In Airways. Biomedical Engineering Society Annual Fall Meeting, Baltimore, MD, September, 2005. 12. Zheng, Y., H. Fujioka, and J.B. Grotberg. Theoretical Study of Effects of Inertia and Gravity on Steady Plug Propagation in a 2-D Channel. Biomedical Engineering Society Annual Fall Meeting, Baltimore, MD, September, 2005. 13. Fujioka, H., D. Halpern, and J.B. Grotberg. Effect of oscillatory core-flow on a shear-thinning fluid layer coating the inner surface of a tube. American Physical Society, Division of Fluid Dynamics, 58th Annual Meeting, Chicago, IL, November, 2005. 14. Zheng, Y., Fujioka, H. and Grotberg, J.B. Effects of Inertia and Gravity on Steady Plug Propagation in a 2-D Channel, 5th World Congress of Biomechanics, Munich, Germany, July 29-Augest 4, 2006. 15. Y. Zheng, H. Fujioka, J.B. Grotberg. Effects of Inertia, Gravity and Surfactant on Steady Plug Propagation in a 2D Channel, Biomedical Engineering Society, Chicago, Illinois, October 11-14, 2006. 16. Fujioka, H. Halpern, D., and Grotberg, J. B. Effect of oscillatory air-flow on the mucus fluid layer coating the inner surface of pulmonary airways, Biomedical Engineering Society, Annual Fall Meeting, Chicago, Illinois, October 11-14, 2006. 17. Fujioka, H. and Grotberg, J. B., Propagation of surfactant-laden liquid plug in liquid lined tube, 3rd International Conference of The International Marangoni Association, University of Florida, USA, November 16, 2006. 18. Fujioka, H., Halpern, D., and Grotberg, J.B. Effect of parenchymal tethering on the Steady propagation of a liquid plug in a Flexible airway model, ASME Summer Bioengineering Conference (SBC2007), Keystone Resort & Conference Center, Keystone, Colorado, USA, June 22, 2007.

Bibliography Type:	Description: (Last Updated: 09/14/2007)
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